FAULT CODE 2215 - Fuel Pump Delivery Pressure - Data Valid But Below Normal Operating Range - Moderately Severe Level

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check if engine will **not** start or engine starts and dies.	
	STEP 1A. Attempt to start the engine.	Engine starts and continues running?
STEP 2.	Check the low pressure side of the fuel system.	
	STEP 2A. Check for external fuel leaks.	Fuel leaking externally?
	STEP 2B. Check for air in the fuel.	Air present in the fuel flow?
	STEP 2C. Check the fuel supply pressure sensor.	Fuel inlet restriction greater than 102 mm-Hg [4 in-Hg]?
	STEP 2D. Inspect the original equipment manufacturer (OEM) fuel supply hose and fuel tank.	Stage 1 filter restriction greater than specification?
	STEP 2E. Check the Stage 1 filter restriction.	Stage 1 filter restriction less than specification?
STEP 3.	Clear the fault codes.	
	STEP 3A. Disable the fault code.	Fault Code 2215 inactive?
	STEP 3B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check if engine will **not** start or engine starts and dies.

STEP 1A. Attempt to start the engine.

Conditions :
- Start engine.

Action	Specification/Repair	Next Step
Try to start the engine. Check if engine will **not** start or engine starts and stalls.	Engine starts and continues running? **YES**	**2A**
	Engine starts and continues running? **NO**	**Reference the Engine Performance TT Symptom tree**

STEP 2. Check the low pressure side of the fuel system.

STEP 2A. Check for external fuel leaks.

Conditions :
- Operate engine.

Action	Specification/Repair	Next Step

Check for external fuel leaks. • Check for external fuel leaks or evidence of leaks.	Fuel leaking externally? **YES** **Repair :** Repair all fuel leaks. • Use the following procedure in the QSK19, QSK19 CM850 Modular Common Rail System, and QSK19 CM2150 MCRS Service Manual, Bulletin 4021592. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html)	**3A**
	Fuel leaking externally? **NO**	**2B**

STEP 2B. Check for air in fuel.

Conditions :

- Remove air bleed line from air bleed valve on the fuel drain manifold block.
- Route the air bleed line into a suitable container to collect fuel.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step

Inspect the fuel flow for air. Use the following procedure in Service Manual, K38, K50, QSK38 and QSK50, Bulletin 4021528. Refer to Procedure 006-003 in Section 6. (/qs3/pubsys2/xml/en/procedures/28/28-006-003.html) Use the following procedure in Service Manual, QSK45 and QSK60, Bulletin 4021530. Refer to Procedure 006-003 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-003.html)	Air present in the fuel flow? **YES** **Repair :** Repair all fuel leaks. • Use the following procedure in the QSK19, QSK19 CM850 Modular Common Rail System, and QSK19 CM2150 Modular Common Rail System Service Manual, Bulletin 4021592. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-024-tr.html)	**3A**
	Air present in the fuel flow? **NO**	**2C**

STEP 2C. Check the fuel supply pressure sensor.

Conditions :

- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Check the fuel supply pressure sensor accuracy. Start and operate the engine at high idle. Record the fuel supply sensor reading in INSITE™ electronic service tool. Shut the engine down and remove the fuel supply pressure sensor. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html) Install a pressure gauge. Start and operate the engine at high idle. Record the fuel supply pressure reading on a pressure gauge installed in place of the fuel supply pressure sensor.	INSITE™ electronic service tool and pressure gauge readings within 14 kPa [2 psia] of each other? **YES** **Repair :** Install the fuel supply pressure sensor that was removed. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)	**2D**
	INSITE™ electronic service tool and pressure gauge readings within 14 kPa [2 psia] of each other? **NO** **Repair :** Replace the fuel supply pressure sensor. Refer to Procedure 019-398 in Section 19. (/qs3/pubsys2/xml/en/procedures/05/05-019-398.html)	**3A**

STEP 2D. Inspect the OEM fuel supply hose and fuel tank.

Conditions :

- Turn keyswitch OFF.

Action	**Specification/Repair**	**Next Step**

Measure fuel inlet restriction at the Stage 1 fuel filter inlet. • Use the following procedure in the QSK19, QSK19 CM850 Modular Common Rail System and QSK19 CM2150 Modular Common Rail System Service Manual, Bulletin 4021592. Refer to Procedure 006-020 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-020-tr.html)	Stage 1 inlet restriction greater than specification? **YES** **Repair :** Refer to the equipment manufacturer service information for repair instructions.	**3A**
	Stage 1 inlet restriction greater than specification? **NO**	**2E**

STEP 2E. Check the Stage 1 filter restriction.

Conditions :

- Turn keyswitch OFF.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Measure the Stage 1 fuel filter restriction. • Use the following procedure in the QSK19, QSK19 CM850 Modular Common Rail System and QSK19 CM2150 Modular Common Rail System Service Manual, Bulletin 4021592. Refer to Procedure 006-020 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-020-tr.html)	Stage 1 filter restriction less than specification? **YES** **Repair :** Replace the high-pressure pump assembly. • Use the following procedure in the QSK19, QSK19 CM850 Modular Common Rail System and QSK19 CM2150 Modular Common Rail System Service Manual, Bulletin 4021592. Refer to Procedure 005-016 in Section 5. (/qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html)	**3A**
	Stage 1 filter restriction less than specification? **NO** **Repair :** Replace the Stage 1 fuel filter. • Use the following procedure in the QSK19, QSK19 CM850 Modular Common Rail System and QSK19 CM2150 Modular Common Rail System Service Manual, Bulletin 4021592. Refer to Procedure 006-075 in Section 6. (/qs3/pubsys2/xml/en/procedures/20/20-006-075-tr.html)	**3A**

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault code.

Conditions :
- Connect all components
- Turn keyswitch ON
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable the fault code. • Use INSITE™ electronic service tool to verify that the fault code is inactive.	Fault Code 2215 inactive? **YES**	**3B**
	Fault Code 2215 inactive? **NO** **Repair :** Return to the troubleshooting steps or contact a Cummins® Authorized Repair location if all steps have been completed and checked a second time.	**1A**

STEP 3B. Clear the inactive fault codes.

Conditions :
- Connect all components
- Turn keyswitch ON
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? **YES**	**Repair complete**
	All fault codes cleared? **NO** **Repair :** Troubleshoot any remaining active fault codes.	**Appropriate troubleshooting steps**

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